

Week 2: Data Sources

From BBG: we got historical spinoff data for S&P 500 from 01/01/2010 - 06/16/2026

From WRDS:

We got daily price data from CRSP

And daily index constituents from Compustat as well as firm fundamentals

Week 3: EDA and Statistical Significance

- Prices + volume: We got prices for both the parent company and the spun off company after the spin-off completed and did some EDA on abnormal returns following this period
- We got data on passive ETF aum that tracks S&P 500 and computed a prediction of forced flow for each spun off company
- Found a positive relationship between index weight and cumulative abnormal return
- Large lead time between announcement of spinoff and actual ex-date (median of 333 calendar days)

Group	Children	Implication
Immediately added to index	CARR, OTIS, VTRS, CEG, WBD, GEHC, VLT0, SOLV, GEV	Passive funds must BUY – no forced selling. These are NOT clean short candidates.
Briefly added then removed	VNT (5 mo), OGN (2.4 yr), AMTM (2.5 mo), FTRE (2 days), CHX (same day)	Mixed signal; short window between effective date and addition to index
Never added to index	ARNC, DTM, SLVM, VMW, KD, ONL, ZIMV, EMBC, ATMU, CHNG	Pure forced-selling pressure from passive funds. Cleanest short candidates.

Post-Completion Spinoff Long: A Risk-Managed Strategy

"Buy every spinoff parent and hold" isn't a strategy — we stress-tested it with a market hedge, a fundamentals screen, and a managed exit to see what actually survives.

THREE RISK LAYERS, TESTED INDEPENDENTLY

Beta hedge

Hedge each position by its **estimated beta** (not a flat 1.0) against the S&P 500 instead of a naive market adjustment.

ADOPTED

Quality screen

Filter by parent leverage + ROA at announcement. Result **reversed** the expected direction — real pattern, but $n=8-9$ per bucket.

UNPROVEN LEAD

Trailing stop

Exit on drawdown from peak CAR before the 12-month mark, tested at 10–25% thresholds.

REJECTED

Portfolio check: up to **10** concurrent positions historically, typically **4–6** — a genuinely diversified book, not sequential single-name bets.

FINAL STRATEGY – HEDGED, 12-MONTH HOLD

MEAN RETURN

+14.7%

MEDIAN RETURN

+21.7%

WIN RATE

75%

SHARPE (PER TRADE)

0.67

SIGNIFICANCE

 $p = 0.007$ $n = 20$ historical spinoffs, 2020–2024

Said upfront, not waited for: worst single outcome was –25% (Ecolab). Sample is small because spinoffs are rare, not because we cut corners. The quality screen is a lead worth tracking, not yet a sizing rule.

STRATEGY

Equal-weight, **beta-hedged** long positions in parent stocks, entered at the spinoff **effective date** (not announcement), held a fixed **12 months**, no dynamic exit.

Pearson r: linear correlation, -1 to +1
 prob_included vs. 252d parent CAR · p =
 0.004

Sizing & classifying the parent long with prob_included

Using inclusion_probability_model's LOO-CV

leave-one-out cross-validation | inclusion probability to size and classify

the beta-hedged parent post-completion long

(CAR = cumulative abnormal return). n = 16 events throughout.

01 WHAT R MEANS, AND WHERE IT SHOWS UP

r = Pearson correlation coefficient: it measures the strength and direction of a straight-line relationship between two variables. **r** = 0 means no linear relationship, **r** = +1 a perfect positive one. Below, **r** is computed between **prob_included** (the model's probability the child gets added to the index) and the parent's CAR at each holding horizon.

horizon	r (naive)	r (β-hedged)	p
21d	0.10	0.10	0.71
63d	-0.01	0.04	0.90
126d	0.26	0.31	0.25
252d	0.64	0.67	0.004

Reading it: at 21d/63d, r is close to 0 — no relationship, and p > 0.7 means it's statistically indistinguishable from noise. At 252d, r = 0.67 is a fairly strong positive correlation, and p = 0.004 says that's very unlikely to be chance. It's the only horizon either approach below relies on.

02 SIZING POSITIONS RAISES THE SHARPE RATIO

$w_i = \text{prob_included}_i / \text{mean}(\text{prob_included})$

w_i = position weight given to event i · prob_included_i = inclusion probability for event i · $\text{mean}(\text{prob_included})$ = average probability across all 16 events, so the average weight stays 1.0x (same total capital as equal-weight, just reallocated).

scheme	mean CAR	σ	Sharpe
equal-weight	15.4%	22.2%	0.69
prob-weighted	23.4%	18.8%	1.25

Sharpe ratio here = mean 252d CAR ÷ σ (its standard deviation), assuming a 0% risk-free rate. Sizing by **prob_included** instead of equal-weighting takes it from **0.69x to 1.25x** in-sample — a real lift, but mechanically expected, since the weights are built straight from the 252d correlation in Part 1. It isn't independent confirmation; the real test is whether it holds on spinoffs outside this sample.

03 CLASSIFICATION TREE: WHAT ACTUALLY DRIVES THE SPLIT

Each event's outcome is bucketed into **terciles** (thirds) of the 252d CAR distribution: underperform (bottom third), middle, outperform (top third). The tree predicts that bucket from 4 inputs:

prob_included

index-inclusion probability

log_child_mktcap

child's market cap, logged

forced_flow_adv

forced-selling flow, in ADV (average daily volume) multiples

beta

parent's market beta

Method: grow the tree fully, then prune it back with cost-complexity pruning (**ccp_alpha** = penalty per extra leaf); pick the pruning level with the best **L00-CV** accuracy — refit on 15 events, test on the 1 left out, repeated for all 16 — instead of guessing a depth by hand.

38%

BASELINE: ALWAYS GUESS
THE MOST COMMON BUCKET

50%

BEST LOO-CV ACCURACY
AT DEPTH 1 (ONE SPLIT)

Learned rule (the one split that generalizes best):

if **forced_flow_adv** (forced-selling flow, ADV mult.) ≤ 7.43 →
 predict "middle"
 else → predict "outperform"

The tree never predicts "underperform" — that bucket is where it's weakest (0% precision and recall on it).